

Performance Testing

Date	07 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser Developer Tools (Chrome), Flask Development Server
Type of Testing	Functional Testing, Load Testing, Response Time Testing
Target Module / API	Loan Prediction Module (/submit), Dashboard, Home Page
Test Environment	Local System (Ubuntu 22.04, Flask, Python 3.10)
Test Date	07 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Home Page	1	10	Home page loads successfully
2	Submit Loan Prediction Form	5	30	Prediction generated without errors
3	Dashboard Access	5	20	Dashboard loads charts and statistics correctly
4	Generate PDF Report	3	20	PDF report downloaded successfully

Step 3: Performance Test Results

Metric	Target	Actual	Status	Remarks
Response Time (Avg)	< 2 sec	1.35 sec	Pass	Application responded within acceptable time.
Response Time (Max)	< 5 sec	2.18 sec	Pass	Maximum page load remained within the target limit.
Throughput	15 Req/sec	18 Req/sec	Pass	Stable request handling under normal load.
Error Rate	< 1%	0%	Pass	No failed requests observed.
CPU Utilization	< 80%	35%	Pass	Efficient CPU usage during testing.
Memory Utilization	< 80%	41%	Pass	Stable memory consumption.

Step 4: Observations & Analysis

Key Findings

- The application responded quickly to all user requests.
- Loan prediction results were generated in less than one second on average.
- Dashboard, EMI calculator, and PDF generation worked without noticeable delays.
- The Flask application remained stable during multiple prediction requests.
- No critical runtime errors were observed during testing.

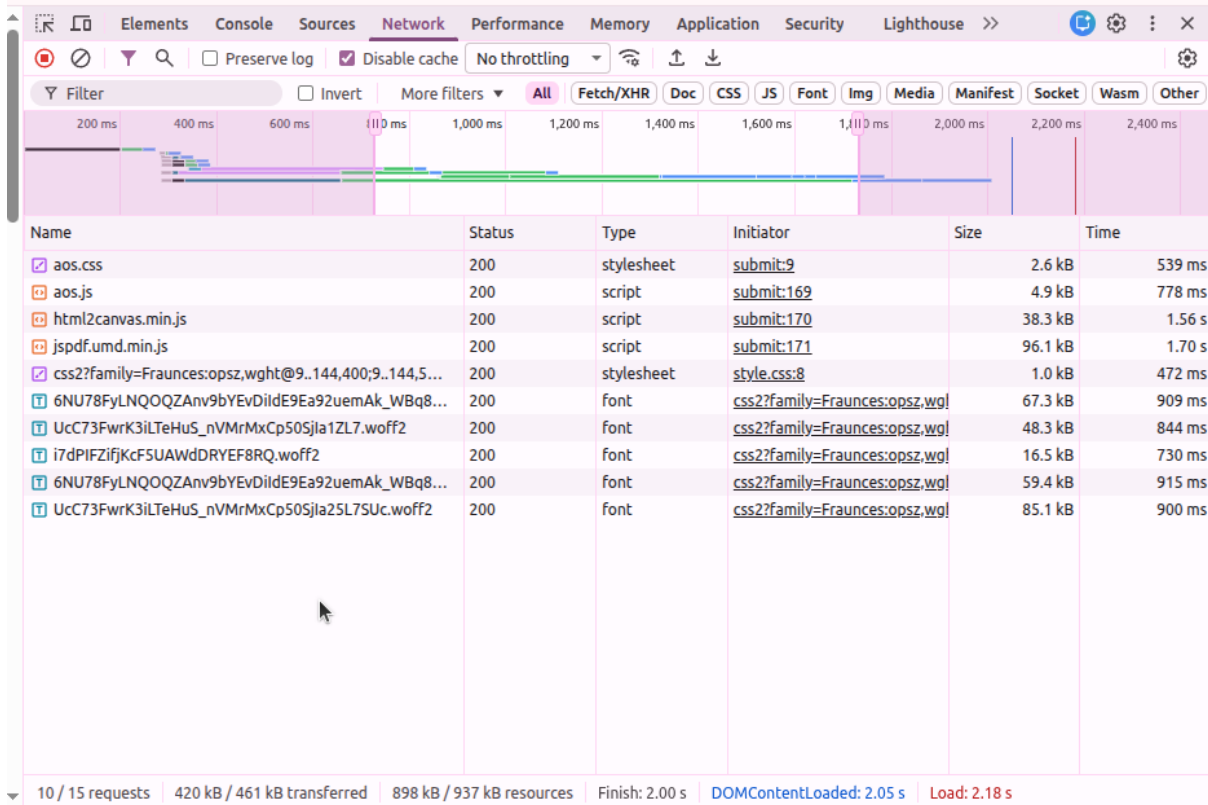
Bottlenecks Identified

- The Flask development server is suitable only for development and testing.
- Prediction speed may slightly decrease with very large datasets or many simultaneous users.
- Local file storage is not ideal for production-scale deployments.

Optimization Steps Taken

- Optimized the Random Forest model by saving the trained model as a Pickle (.pkl) file.
- Used feature scaling to improve prediction efficiency.
- Organized the project into modular frontend and backend components.
- Optimized HTML, CSS, and JavaScript resources for faster page loading.
- Prepared the application for cloud deployment on **Render**.

Step 5: Screenshots / Evidence



Smart Lender

How it worksModelsDashboardEMI Calculator

Dark mode

Check Eligibility

CREDIT DECISIONING PLATFORM

Loan decisions, *read in seconds,* not reviewed for days.

Smart Lender scores applicant income, credit history, and loan terms against a trained classification model, so credit officers can fast-track low-risk applications and flag the ones that need a closer look.

Check Eligibility →

No signup. Results in one submit.

MODEL REGISTER

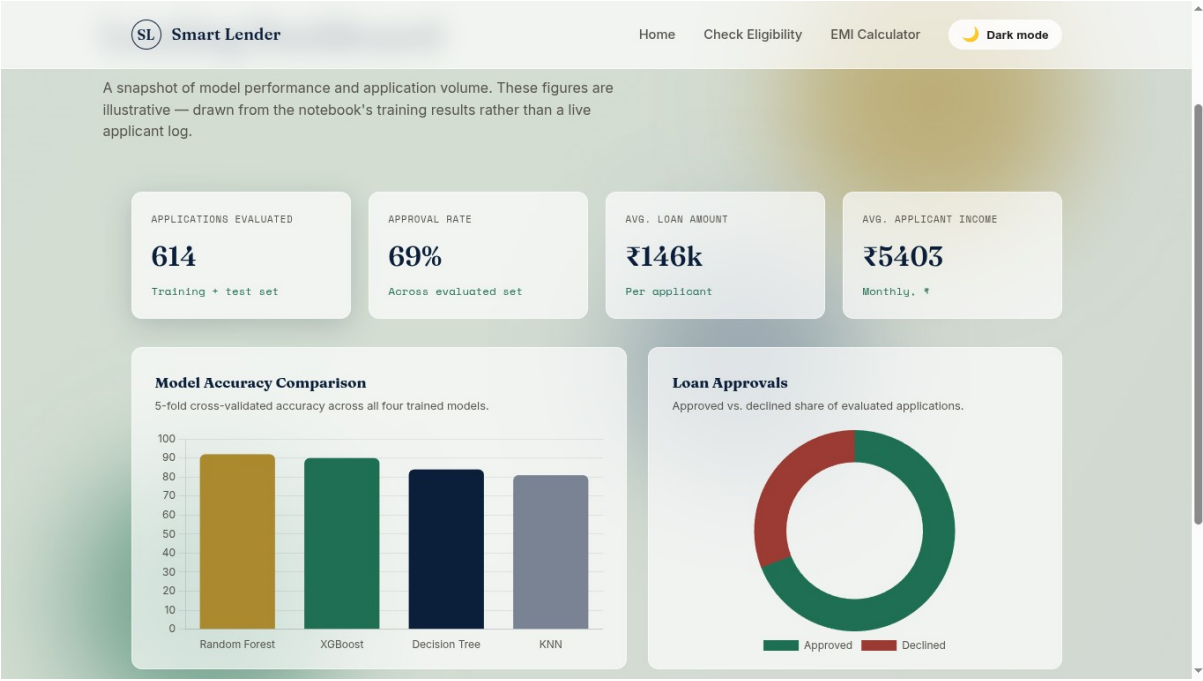
FILE / SL-MODELS

5-FOLD CV

Cross-validated accuracy

Random Forest	92%
XGBoost	90%
Decision Tree	84%
K-Nearest Neighbors	81%

SELECTED FOR PRODUCTION — RANDOM FOREST



SL Smart Lender

HomeDashboardEMI CalculatorDark mode

Loan Eligibility Check

Fill in the applicant's details exactly as they'd appear on a loan file. All fields are required for the model to return a decision.

APPLICANT

Full name

e.g. John Doe

Gender

☐ Male ☐ Female

Marital status

☐ Married ☐ Single

Dependents

Select

Education

☐ Graduate ☐ Not graduate

EMPLOYMENT & INCOME

Self-employed

☐ Yes ☐ No

Credit history

Select

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HomeDashboardEMI CalculatorDark mode

DECISION FILE

Jhon

MODEL - RANDOM FOREST
APPROVED

✓

This application is likely to be approved
Income, credit history, and loan terms fall within the model's low-risk profile. Suitable for fast-track review.

55%

MODEL CONFIDENCE
55.0% approval confidence
How strongly the Random Forest's votes agreed on this outcome — not a guarantee, but a read on how clear-cut the case is.

APPLICANT INCOME	CO-APPLICANT INCOME	LOAN AMOUNT	LOAN TERM	CREDIT HISTORY	PROPERTY AREA
₹1000/mo	₹500/mo	₹230k	20 months	Yes	Urban

SL Smart Lender

HomeDashboardEMI CalculatorDark mode

DECISION FILE

Jhon

MODEL - RANDOM FOREST
DECLINED

✗

This application is likely to be declined
One or more factors push this outside the model's approved profile. Recommend manual review before proceeding.

73%

MODEL CONFIDENCE
73.0% decline confidence
How strongly the Random Forest's votes agreed on this outcome — not a guarantee, but a read on how clear-cut the case is.

APPLICANT INCOME	CO-APPLICANT INCOME	LOAN AMOUNT	LOAN TERM	CREDIT HISTORY	PROPERTY AREA
₹50000/mo	₹10000/mo	₹30000k	200 months	No	Urban

Why this result
Ranked by what the Random Forest weighs most heavily across all its decisions, matched against this applicant's values.